BIOCHEMISTRY CUMULATIVE EXAM
December 5, 1998

1. Glucose labeled with C-14 at carbon atom number 6 is added to a healthy liver cell suspension actively undergoing respiration with adequate supplies of unlabeled metabolites, and then a short period later glutamic acid is isolated and analyzed to determine the first carbon atom(s) in the glutamic acid which become labeled with C-14.

   a) (20) Briefly outline the metabolic pathways needed for the conversion of glucose to glutamic acid under the above conditions.

   b) (35) Trace the C-14 atom on glucose carbon number 6 through the above pathways and identify the first carbon atom(s) on glutamic acid which will be labeled. Show the labeling of key intermediates along the pathway.

   c) (25) Identify at least one key enzyme-catalyzed reaction in the above pathway that determines which specific carbon atom(s) are labeled, and discuss what the observed labeling pattern tells us about the stereochemistry of that reaction at the active site of the enzyme.

2. The above experiment is repeated, except that glucose labeled at carbon 6 is added to a liver cell suspension in a severe state of ketosis which is completely depleted of citric acid cycle intermediates.

   d) (10) Briefly outline how the metabolic pathways needed to convert glucose to glutamic acid would differ from experiment 1.

   e) (10) Identify the first carbon atom(s) in glutamic acid which will be labeled under the conditions of experiment 2.

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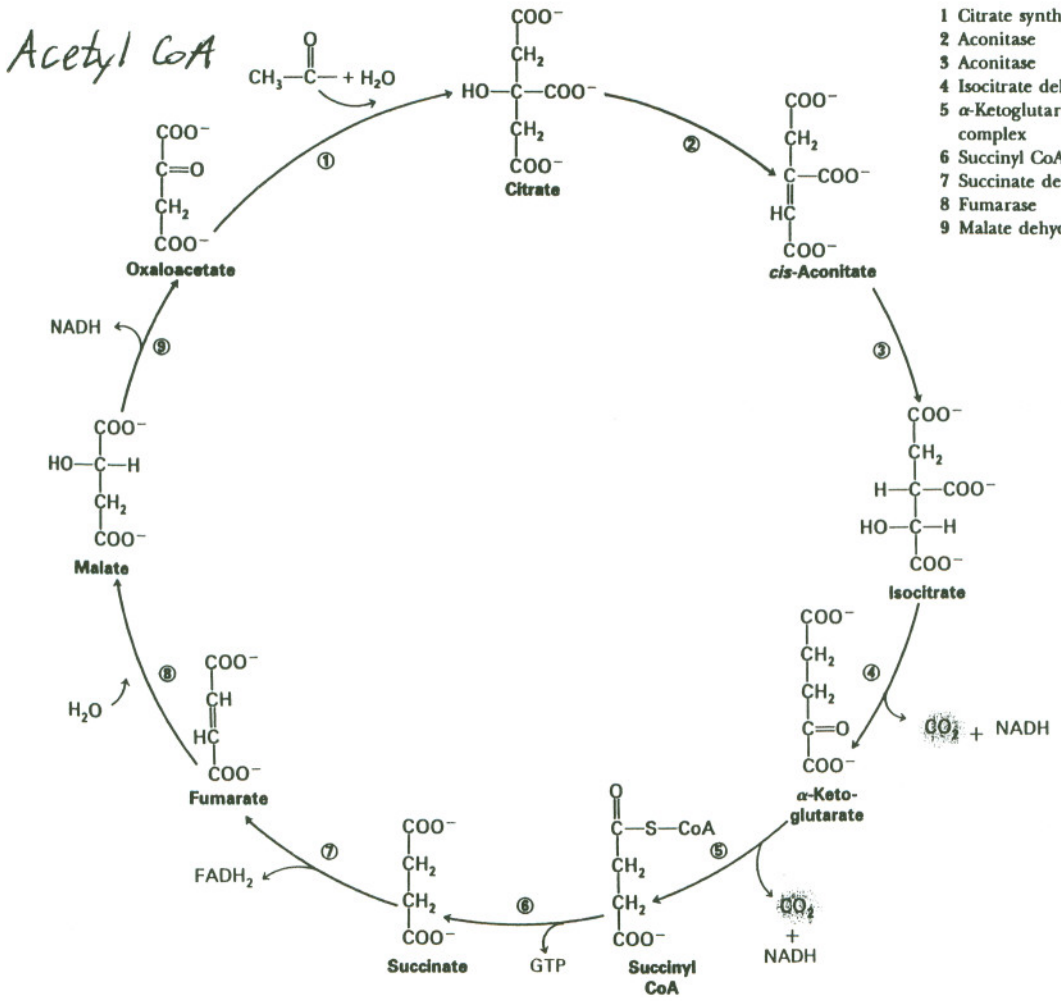
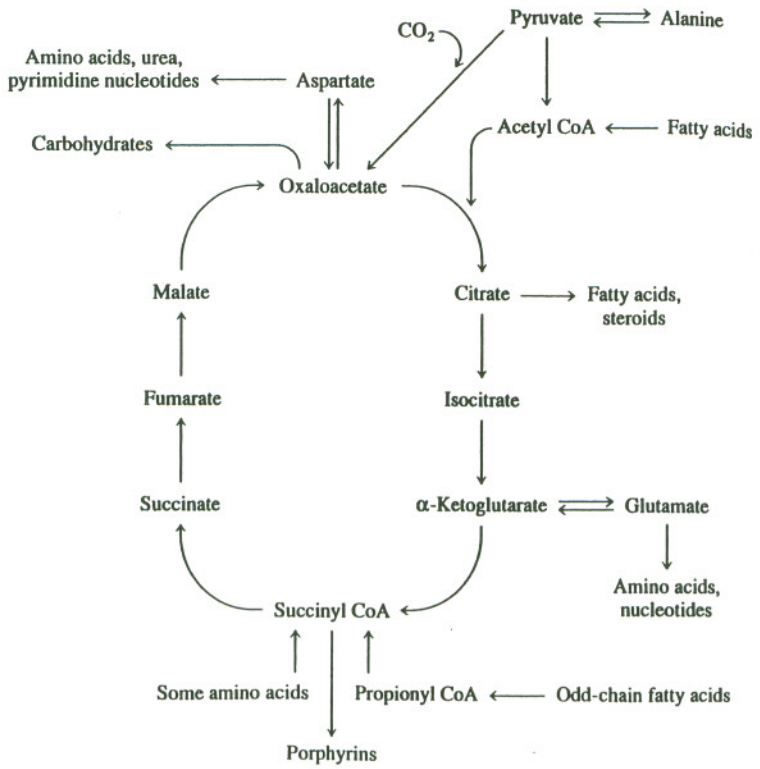
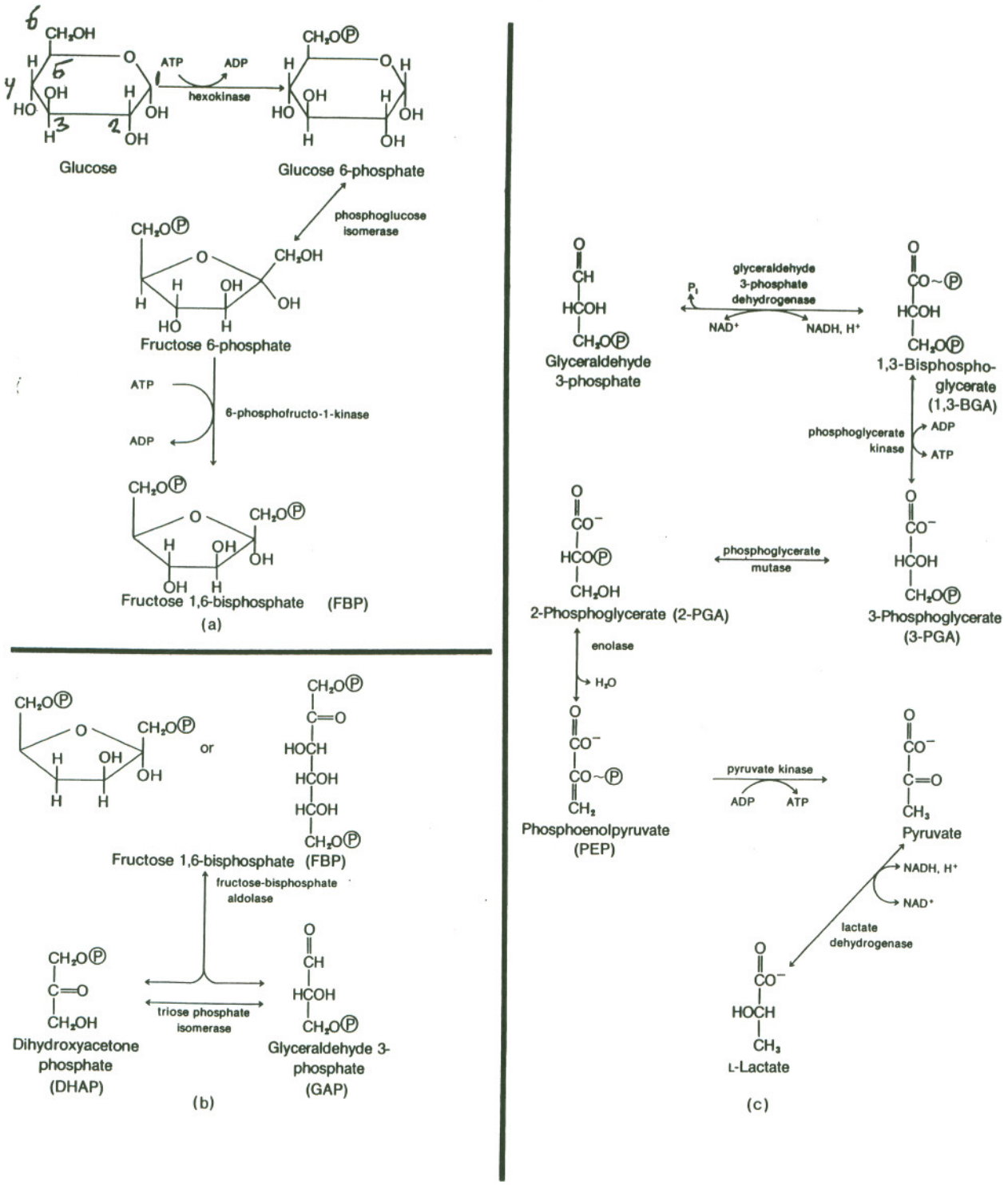
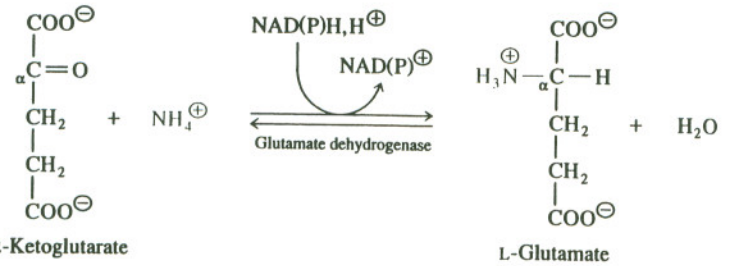
1 Citrate syntha
2 Aconitase
3 Aconitase
4 Isocitrate deh
5 α-Ketoglutara complex
6 Succinyl CoA
7 Succinate deh
8 Fumarase
9 Malate dehyd






Glucose
ATP
ADP
hexokinase
Glucose 6-phosphate
phosphoglucose isomerase
Fructose 6-phosphate
ATP
ADP
6-phosphofructo-1-kinase
Fructose 1,6-bisphosphate (FBP)
(a)
Fructose 1,6-bisphosphate (FBP)
or
fructose-bisphosphate aldolase
triose phosphate isomerase
Dihydroxyacetone phosphate (DHAP)
Glyceraldehyde 3-phosphate (GAP)
(b)
Glyceraldehyde 3-phosphate
glyceraldehyde 3-phosphate dehydrogenase
NAD+
NADH, H+
1,3-Bisphosphoglycerate (1,3-BGA)
phosphoglycerate kinase
ADP
ATP
2-Phosphoglycerate (2-PGA)
phosphoglycerate mutase
3-Phosphoglycerate (3-PGA)
enolase
H2O
Phosphoenolpyruvate (PEP)
pyruvate kinase
ADP
ATP
Pyruvate
NADH, H+
NAD+
lactate dehydrogenase
L-Lactate
(c)


NAD(P)H,H⊕
NAD(P)⊕
Glutamate dehydrogenase
α-Ketoglutarate
L-Glutamate